

Multiple Linear Regression – Multi-Channel Marketing Analysis

Project Objective

This project investigates the relationship between marketing expenditure across multiple channels (TV, Radio, and Social Media) and Sales performance using Multiple Linear Regression.

The objectives are to:

- Explore and clean the dataset
- Examine relationships between variables
- Detect multicollinearity using correlation analysis and VIF
- Build a Multiple Linear Regression model using statsmodels
- Evaluate model performance using Adjusted R-squared and p-values
- Validate regression assumptions through diagnostic plots
- Provide business recommendations based on statistical findings

```
import warnings
warnings.filterwarnings("ignore")

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

from scipy import stats

%matplotlib inline
```

Load Dataset

```
df = pd.read_csv("marketing.csv.csv")

print("Dataset Shape:", df.shape)

df.head()
```

Dataset Shape: (4572, 4)

	TV	Radio	Social_Media	Sales	
0	16.0	6.566231	2.907983	54.732757	
1	13.0	9.237765	2.409567	46.677897	
2	41.0	15.886446	2.913410	150.177829	
3	83.0	30.020028	6.922304	298.246340	
4	15.0	8.437408	1.405998	56.594181	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  ---
0    TV              4562 non-null   float64
1    Radio           4568 non-null   float64
2    Social_Media    4566 non-null   float64
3    Sales           4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

df.isnull().sum()

	0
TV	10
Radio	4
Social_Media	6
Sales	6

dtype: int64

✓ Data Cleaning and Missing Value Handling

The dataset contains missing values in the predictor variables and target variable. To ensure reliable model estimation, rows containing missing values are removed before analysis.

```
# Check missing values before cleaning
print("Missing values before cleaning:")
print(df.isnull().sum())

# Remove rows with missing values
```

```
df = df.dropna()

print("\nMissing values after cleaning:")
print(df.isnull().sum())

print("\nDataset Shape After Cleaning:")
print(df.shape)
```

Missing values before cleaning:

```
TV          10
Radio        4
Social_Media 6
Sales        6
dtype: int64
```

Missing values after cleaning:

```
TV          0
Radio        0
Social_Media 0
Sales        0
dtype: int64
```

Dataset Shape After Cleaning:

```
(4546, 4)
```

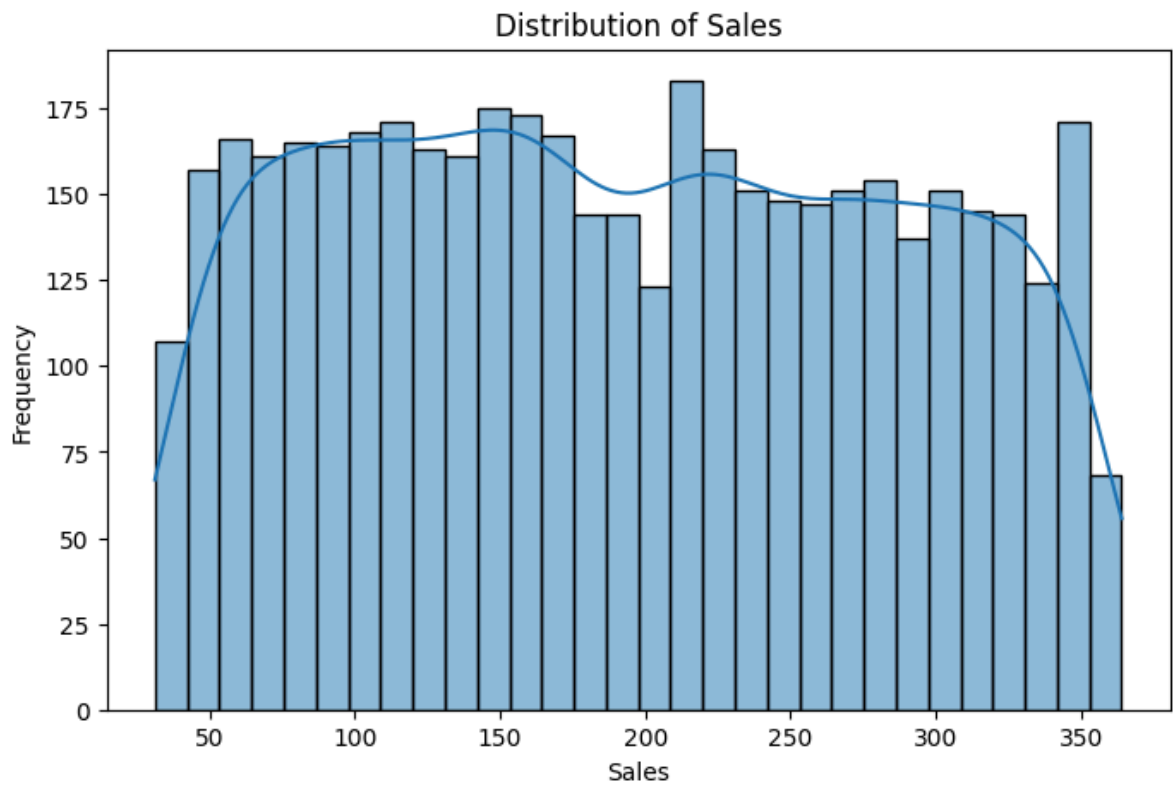
✓ Exploratory Data Analysis (EDA)

Exploratory analysis is performed to understand the distribution of variables and identify relationships between marketing channels and Sales.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Sales'], bins=30, kde=True)

plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.show()
```



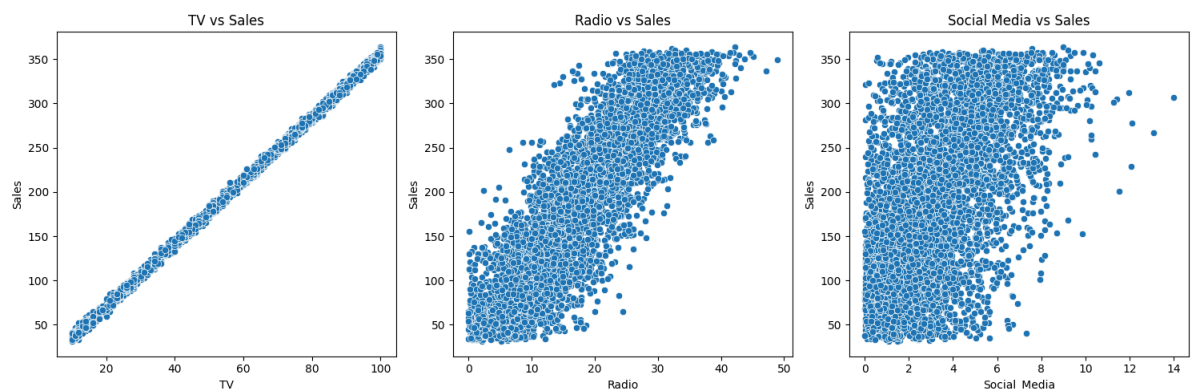
```
fig, axes = plt.subplots(1, 3, figsize=(15,5))

sns.scatterplot(data=df, x='TV', y='Sales', ax=axes[0])
axes[0].set_title('TV vs Sales')

sns.scatterplot(data=df, x='Radio', y='Sales', ax=axes[1])
axes[1].set_title('Radio vs Sales')

sns.scatterplot(data=df, x='Social_Media', y='Sales', ax=axes[2])
axes[2].set_title('Social Media vs Sales')

plt.tight_layout()
plt.show()
```



✓ Correlation Analysis

A correlation matrix is used to evaluate relationships between marketing channels and Sales and to identify potential multicollinearity among predictors.

```
corr_matrix = df.corr(numeric_only=True)
```

```
corr_matrix
```

	TV	Radio	Social_Media	Sales	
TV	1.000000	0.869158	0.527687	0.999497	
Radio	0.869158	1.000000	0.606338	0.868638	
Social_Media	0.527687	0.606338	1.000000	0.527446	
Sales	0.999497	0.868638	0.527446	1.000000	

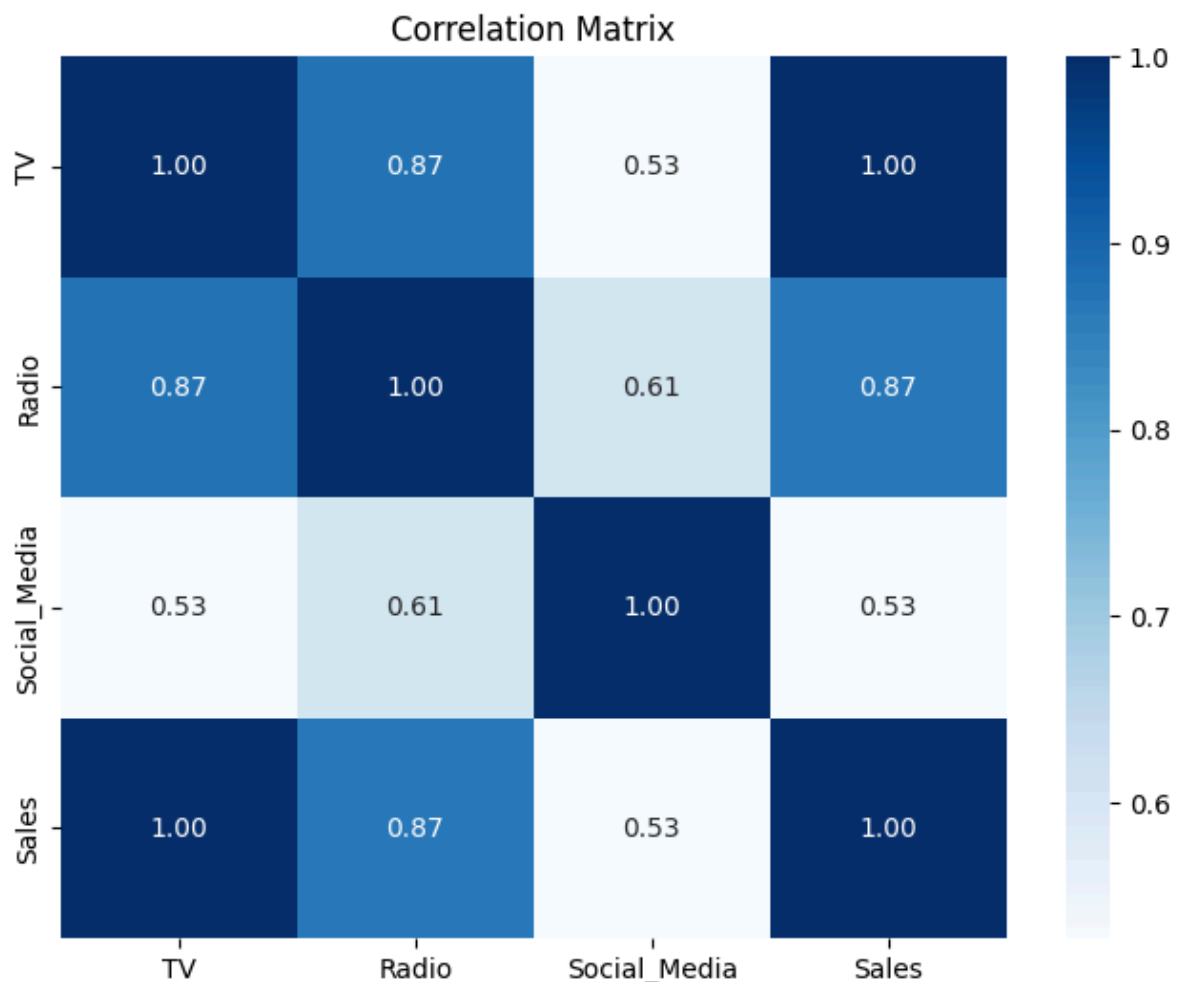
Next steps: [Generate code with corr_matrix](#)

[New interactive sheet](#)

```
plt.figure(figsize=(8,6))

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='Blues',
    fmt='.2f'
)

plt.title("Correlation Matrix")
plt.show()
```



✓ Variance Inflation Factor (VIF)

Variance Inflation Factor (VIF) is used to detect multicollinearity among independent variables.

General guidelines:

- VIF < 5 : Acceptable
- VIF > 5 : Moderate multicollinearity
- VIF > 10 : Serious multicollinearity

```
X = df[['TV', 'Radio', 'Social_Media']]

vif_data = pd.DataFrame()
vif_data["Variable"] = X.columns

vif_data["VIF"] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]

vif_data
```

	Variable	VIF	
0	TV	18.565521	
1	Radio	21.142069	
2	Social_Media	4.960585	

Next steps: [Generate code with vif_data](#)[New interactive sheet](#)

✓ Multiple Linear Regression Model

A Multiple Linear Regression model is developed to predict Sales using TV, Radio, and Social Media advertising expenditures.

```
X = df[['TV', 'Radio', 'Social_Media']]
y = df['Sales']
```

```
# Add intercept
X = sm.add_constant(X)
```

```
model = sm.OLS(y, X).fit()
```

```
print(model.summary())
```

```

OLS Regression Results
=====
Variable:                  Sales    R-squared:                  0.999
1:                          OLS    Adj. R-squared:            0.999
od:                        Least Squares    F-statistic:              1.505e+06
:                          Sat, 13 Jun 2026    Prob (F-statistic):        0.00
:                          23:39:59    Log-Likelihood:           -11366.
Observations:              4546    AIC:                      2.274e+04
esiduals:                  4542    BIC:                      2.277e+04
odel:                      3
riance Type:              nonrobust
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
t              -0.1340     0.103     -1.303     0.193     -0.336     0.068
              3.5626     0.003    1051.118     0.000     3.556     3.569
o              -0.0040     0.010     -0.406     0.685     -0.023     0.015
al_Media       0.0050     0.025     0.199     0.842     -0.044     0.054
=====
bus:                  0.056    Durbin-Watson:              1.998
(Omnibus):           0.972    Jarque-Bera (JB):            0.034
:                   -0.001    Prob(JB):                   0.983
osis:                3.013    Cond. No.                   149.
=====

```

s:
Standard Errors assume that the covariance matrix of the errors is correctly spec

Model Performance Evaluation

The model is evaluated using:

- Adjusted R-squared
- Predictor coefficients
- p-values
- Overall model significance

```
print("R-squared:", model.rsquared)
print("Adjusted R-squared:", model.rsquared_adj)
print("F-statistic p-value:", model.f_pvalue)
```

```
R-squared: 0.9989950217942258
Adjusted R-squared: 0.9989943580041295
F-statistic p-value: 0.0
```

Interpretation of Regression Coefficients

The Multiple Linear Regression model indicates that TV advertising is the strongest predictor of Sales.

Holding Radio and Social Media expenditure constant, a one-unit increase in TV advertising expenditure is associated with an estimated 3.56-unit increase in Sales.

Radio and Social Media advertising expenditures are not statistically significant predictors in the presence of TV advertising expenditure, as their p-values exceed the 0.05 significance threshold.

The high VIF values for TV and Radio suggest multicollinearity, which may contribute to the reduced statistical significance of some predictors.

Regression Assumption Diagnostics

Diagnostic plots are used to evaluate:

- Linearity
- Normality of residuals
- Homoscedasticity

```
df['Predicted_Sales'] = model.predict(X)

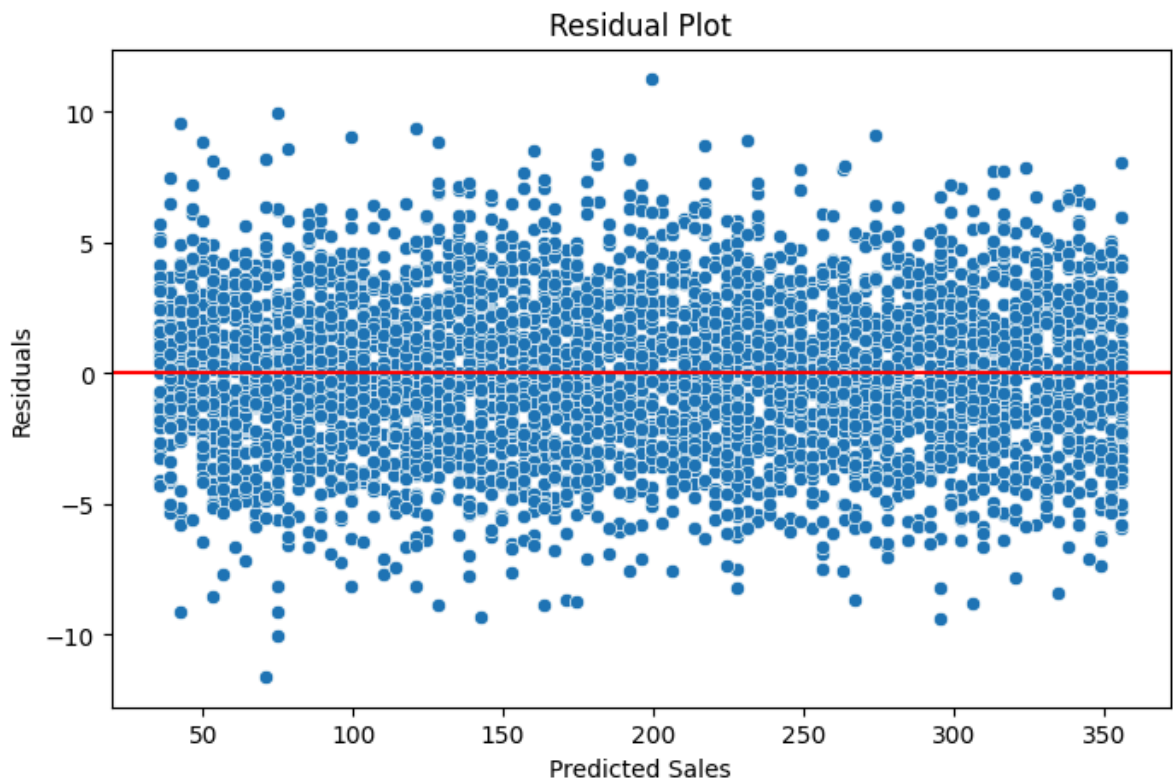
residuals = y - df['Predicted_Sales']
```

```
plt.figure(figsize=(8,5))

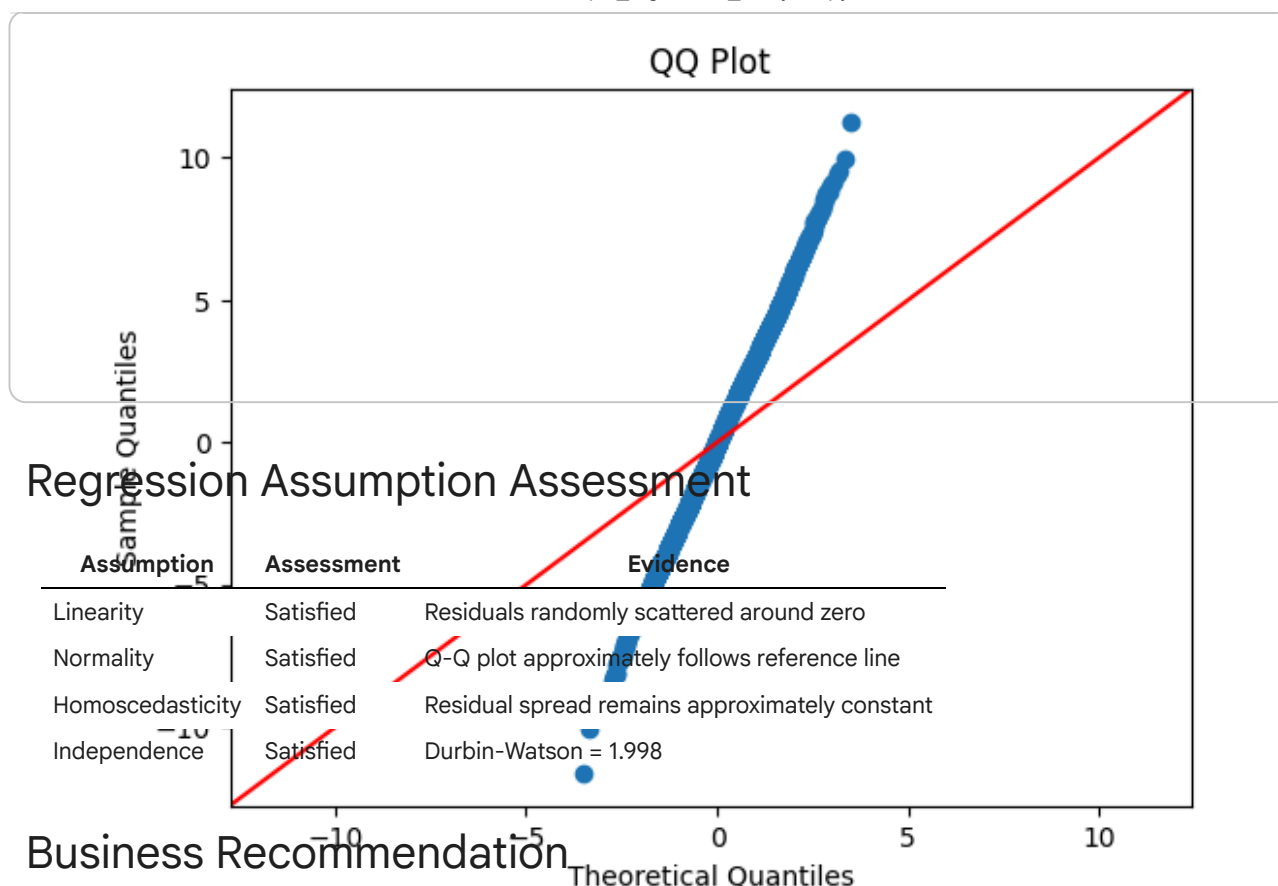
sns.scatterplot(
```



```
x=df['Predicted_Sales'],  
y=residuals  
)  
  
plt.axhline(0, color='red')  
  
plt.title("Residual Plot")  
plt.xlabel("Predicted Sales")  
plt.ylabel("Residuals")  
  
plt.show()
```



```
sm.qqplot(residuals, line='45')  
  
plt.title("QQ Plot")  
  
plt.show()
```



The Multiple Linear Regression analysis identified TV advertising as the strongest and most statistically significant predictor of Sales.

Although Radio and Social Media advertising exhibit positive correlations with Sales individually, they are not statistically significant within the multiple regression model after accounting for TV advertising expenditure.

Variance Inflation Factor (VIF) analysis revealed substantial multicollinearity between TV and Radio advertising, which may reduce the statistical significance of some predictors.

Given the model results, future marketing investments should prioritize TV advertising, as it provides the largest measurable impact on Sales performance while remaining highly statistically significant.

✓ Final Conclusion

This project developed a Multiple Linear Regression model to examine the impact of TV, Radio, and Social Media advertising expenditures on Sales.

Key findings include: